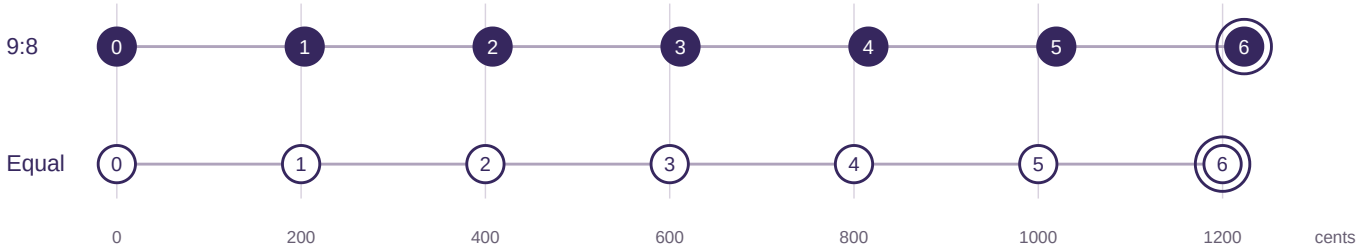
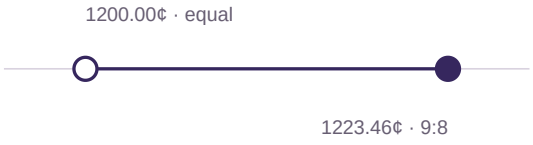
Six tones and an octave

Selected step 6 (H), modern starting reference 220 Hz. Selected sound: Selected pair, together. The audio is available as a separate WAV.



Purple: successive 9:8 tones. Outlined: six equal divisions of a 2:1 octave. Numbers count intervals from zero.

Selected pair · enlarged scale



9:8 route: **446.003036 Hz**; equal route: **440.000000 Hz**.

Difference: 23.460010 cents and 6.003036 Hz. Simultaneous sine tones give this many amplitude beats per second in the generated signal; the control uses two identical equal-route tones.

Seven numbers, six multiplications

A–H label the proof’s numbers. They form a progression, each 9:8 of its predecessor. We realize their ratios as increasing frequencies. They are not labeled as string lengths. Frequencies and cents below are rounded modern calculations.

Menge 1916, printed p. 171, proposition IX; selected row highlighted

Step / letter	Source integer	Exact ratio to A	9:8 Hz	Equal Hz	Difference (¢)
0 / A	262144	1/1	220.000000	220.000000	0.000000
1 / B	294912	9/8	247.500000	246.941651	3.910002
2 / Γ	331776	81/64	278.437500	277.182631	7.820003
3 / Δ	373248	729/512	313.242188	311.126984	11.730005
4 / E	419904	6561/4096	352.397461	349.228231	15.640007
5 / Z	472392	59049/32768	396.447144	391.995436	19.550009
6 / H	531441	531441/262144	446.003036	440.000000	23.460010

531441 > 2 × 262144 = 524288. The excess is 7153. Dividing by the octave gives 531441/524288, or 23.460010 cents, commonly called the Pythagorean comma.

Six modern 200-cent steps close the octave by changing each step to 2^(1/6), approximately 1.122462. The source's 9/8 is 1.125. Modern algebra also gives the same comma from twelve 3:2 fifths compared with seven octaves: (3/2)^12 ÷ 2^7 = 531441/524288. This connection is an explanation added here, not a quotation from IX.

Read proposition XIV

The work is transmitted under Euclid's name; authorship is not settled here. Greek text and Latin translation: Heinrich Menge, *Euclidis opera omnia* VIII (Leipzig, 1916), pp. 174–175. English translations and notes are new. The selected app sentence was 3; every sentence is included below.

The claim · XIV

Τὸ διὰ πασῶν ἔλαττον ἢ ἓξ τόνων.

diapason minus est quam sex tonorum.

The octave is smaller than six tones.

ἔλαττον / minus means ‘smaller’; ἢ / quam introduces the comparison. ‘Tone’ here has the 9:8 ratio established in XIII. A modern 200-cent whole tone changes that premise.

The musical premises · XII and XIII

δέδεικται γὰρ τὸ μὲν διὰ πασῶν διπλάσιον, ὁ δὲ τόνος ἐπόγδοος·

nam demonstratum est, diapason duplum esse [prop. XII], tonum vero sesqueoctavum [prop. XIII].

For it has been shown that the octave is double, while the tone is a whole and an eighth.

δέδεικται / demonstratum est points back to earlier proofs. διπλάσιον / duplum names 2:1; ἐπόγδοος / sesqueoctavum names 9:8. Menge adds the proposition references in square brackets to the Latin.

The numerical premise · IX

τὰ δὲ ἓξ ἐπόγδοα διαστήματα μείζονα διαστήματός [ἐστι] διπλασίου.

verum sex intervalla sesqueoctava intervallo duplo sunt maiora [prop. IX].

But six intervals of a whole and an eighth are greater than a double interval.

μείζονα / maiora means ‘greater’. The seven source integers supply six successive ratios, each 9:8. The Greek bracketed [ἐστι] is retained from this edition; it is not silently promoted to certain text.

The conclusion · XIV

τὸ ἄρα διὰ πασῶν ἔλαττόν ἐστιν ἓξ τόνων.

ergo diapason minus est sex tonis.

Therefore the octave is smaller than six tones.

ἄρα / ergo carries the inference. Greek ἓξ τόνων and Latin sex tonis express the comparison. The numerical contradiction belongs to these ratios; it does not make modern equal temperament impossible.

Four words in the argument

διὰ πασῶν · dia pasōn · diapason · octave

Literally ‘through all’. Here it names the consonance with ratio 2:1. The phrase is quoted in its printed form, not translated as ‘all notes’ in the proof.

τόνος · tonos · tonus · tone

The tone in XIII is the interval left when a 4:3 fourth is removed from a 3:2 fifth: (3/2) ÷ (4/3) = 9/8. Removing a musical interval divides ratios; it does not subtract their numerical values.

ἐπόγδοος · epogdoos · sesqueoctavus · a whole and an eighth

A quantity contains the other plus one eighth of it: 1 + 1/8 = 9/8. Each selected source row after A adds one eighth of the preceding integer.

διπλάσιον · diplasion · duplum · double

An octave doubles frequency in this modern realization. At step six the source's target is twice 262144, or 524288. Compare that with H = 531441.

Open this comparison · Complete scanned edition · Code and source credits.

Ratios multiply and cents add. Equal octave divisions are a modern comparison; no ancient absolute pitch, performance or instrument timbre is reconstructed. The six complete source pages and Greek font below are embedded in this document. It opens and prints offline.

ϑ'.

Τὰ ἑξ ἐπόγδοα διαστήματα μείζονά ἐστι διαστήματος ἑνὸς διπλασίου.

ἔστω γὰρ εἷς ἀριθμὸς ὁ *A*. καὶ τοῦ μὲν *A* ἐπόγδοος ἔστω ὁ *B*, τοῦ δὲ *B* ἐπόγδοος ὁ *Γ*, τοῦ δὲ *Γ* ἐπόγδοος ὁ *Δ*, τοῦ δὲ *Δ* ἐπόγδοος ὁ *E*, τοῦ *E* ἐπόγδοος ὁ *Z*, τοῦ *Z* ἐπόγδοος ὁ *H*· λέγω, ὅτι ὁ *H* τοῦ *A* μείζων ἐστὶν ἢ διπλάσιος.

ἐπεὶ ἐμάθομεν εὑρεῖν ἑπτὰ ἀριθμοὺς ἐπογδόους ἀλλήλων, εὑρήσθωσαν οἱ *A*, *B*, *Γ*, *Δ*, *E*, *Z*, *H*, καὶ γίνεται
ὁ μὲν *A* ͵κϛ μύρια ͵βρμδ,
ὁ δὲ *B* ͵κθ μύρια ͵δ͵ψιβ,
ὁ δὲ *Γ* ͵λγ μύρια ͵αψοϛ,
ὁ δὲ *Δ* ͵λξ μύρια ͵γσμη,
ὁ δὲ *E* ͵μα μύρια ͵θ͵ψδ,
ὁ δὲ *Z* ͵μξ μύρια ͵βτ͵ηβ,
ὁ δὲ *H* ͵νγ μύρια ͵αυμα, καί ἐστιν ὁ *H* τοῦ *A* μείζων ἢ διπλάσιος.

ι'.

Τὸ διὰ πασῶν διάστημά ἐστι πολλαπλάσιον.

ἔστω γὰρ νήτη μὲν ὑπερβολαίων ὁ *A*, μέση δὲ ὁ *B*, προσλαμβανόμενος δὲ ὁ *Γ*. τὸ ἄρα *A Γ* διάστημα δὶς διὰ πασῶν ὂν ἐστι σύμφωνον. ἤτοι οὖν ἐπιμόριόν ἐστιν ἢ πολλαπλάσιον. ἐπιμόριον μὲν οὐκ ἔστιν· ἐπιμορίου γὰρ διαστήματος μέσος οὐδεὶς ἀνάλογον ἐμπίπτει· πολλαπλάσιον ἄρα ἐστίν. ἐπεὶ οὖν δύο ⟨ἴσα⟩ διαστήματα τὰ *AB*, *BΓ* συντεθέντα ποιεῖ πολλαπλάσιον τὸ ὅλον, καὶ τὸ *AB* ἄρα ἐστὶ πολλαπλάσιον.

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2. ἐστι] εἰσι W. 9. Post ἀριθμούς add. ἐφεξῆς M³.
11. $\overset{\varkappa}{M}\overset{\varsigma}{}$ codd., item in seqq. (κϛ M²). 12. ͵δ] ϟ M; item lin. 15, τ W. 13. λγ et ͵α add. M². 14. Numeros add. M². 15. μα] M². ͵δ] ψ W. ͵αυ] M². 18. διπλάσιος] α|β|γ|δ|ε|ζ|η add M¹, ῖα ῖβ ῖγ ῖδ ῖε ῖζ ῖη W. 23. ὅν] M⁴, ὧν M¹. ἤ] ἡ M. 26. ἴσα] Meibom, om. codd. τά] om. W. 28. ἐστί — p. 178 lin. 10 τρίτη] om. W; mg. m. 2 λείπει ὧδε.

Menge 1916, printed p. 170; Greek text. Public-domain edition, Internet Archive California scan, PDF p. 228.

IX.

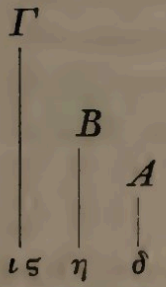
Sex intervalla sesqueoctava maiora sunt uno intervallo duplo.

nam sit unus numerus A. et numero A octava parte maior sit B, numero autem B octava parte maior Γ, numero Γ octava parte maior Δ, numero Δ octava parte maior E, numero E octava parte maior Z, numero Z octava parte maior H. dico, esse $H > 2A$.

quoniam didicimus septem invenire numeros octonis inter se partibus maiores, inventi sint numeri A, B, Γ, Δ, E, Z, H, et fit $A = 262144$;

B 294912,
Γ 331776,
Δ 373248,
E 419904,
Z 472392,
H 531441, et est $H > 2A$.

X.



Diapason intervallum est multiplex.

nam sit nete hyperbolaeon A, mese autem B, proslambanomenos vero Γ. $A\Gamma$ igitur intervallum, quod est bis diapason, consonans est. itaque aut superparticulare est aut multiplex. superparticulare non est; nam superparticularis intervalli nullus medius numerus proportionaliter incidit [prop. III]; quare multiplex est. iam quoniam duo intervalla aequalia AB, $B\Gamma$ composita totum efficiunt multiplex [prop. II], etiam AB multiplex est.

---

9. Heiberg l. c. comparat El. VIII, 2: *ἀριθμοὺς εὑρεῖν ἐξῆς ἀνάλογον ἐλαχίστους, ὅσους ἂν ἐπιτάξῃ τις, ἐν τῷ δοθέντι λόγῳ.*
18. Fig. $\alpha \mid \beta \mid \gamma \mid \delta \mid \varepsilon \mid \zeta \mid \eta$ M^1.
20. Fig. habet N, sine numeris M^2.

Menge 1916, printed p. 171; Latin translation. Public-domain edition, Internet Archive California scan, PDF p. 229.

ια΄.

Τὸ διὰ τεσσάρων διάστημα καὶ τὸ διὰ πέντε ἑκάτερον ἐπιμόριόν ἐστιν.

ἔστω γὰρ νήτη μὲν συνημμένων ὁ A, μέση δὲ ὁ B, ὑπάτη δὲ μέσων ὁ Γ. τὸ ἄρα $A\Gamma$ διάστημα δὶς διὰ τεσσάρων ὄν ἐστι διάφωνον· οὐκ ἄρα ἐστὶ πολλαπλάσιον. ἐπεὶ οὖν δύο διαστήματα ἴσα τὰ AB, $B\Gamma$ συντεθέντα τὸ ὅλον μὴ ποιεῖ πολλαπλάσιον, οὐδὲ ἄρα τὸ AB ἐστι πολλαπλάσιον. καί ἐστι σύμφωνον· ἐπιμόριον ἄρα. ἡ αὐτὴ δὲ ἀπόδειξις καὶ ἐπὶ τοῦ διὰ πέντε.

ιβ΄.

Τὸ διὰ πασῶν διάστημά ἐστι διπλάσιον.

ἐδείξαμεν γὰρ αὐτὸ πολλαπλάσιον. οὐκοῦν ἤτοι διπλάσιόν ἐστιν — ἢ μεῖζον ἢ διπλάσιον. ἀλλ᾽ ἐπεὶ ἐδείξαμεν τὸ διπλάσιον διάστημα ἐκ δύο τῶν μεγίστων ἐπιμορίων συγκείμενον, ὥστε, εἰ ἔσται τὸ διὰ πασῶν μεῖζον διπλασίου, οὐ συγκείσεται ἐκ δύο μόνων ἐπιμορίων, ἀλλ᾽ ἐκ πλειόνων, — σύγκειται δὲ ἐκ δύο συμφώνων διαστημάτων, ἔκ τε τοῦ διὰ πέντε καὶ τοῦ διὰ τεσσάρων, οὐκ ἄρα ἔσται τὸ διὰ πασῶν μεῖζον διπλασίου. διπλάσιον ἄρα.

ἀλλ᾽ ἐπειδὴ τὸ διὰ πασῶν ἐστι διπλάσιον, τὸ δὲ διπλάσιον ἐκ τῶν μεγίστων ἐπιμορίων δύο συνέστηκε, καὶ τὸ διὰ πασῶν ἄρα ἐξ ἡμιολίου καὶ ἐπιτρίτου συνέστηκε· ταῦτα γὰρ μέγιστα. συνέστηκε δὲ ἐκ τοῦ διὰ πέντε καὶ ἐκ τοῦ διὰ τεσσάρων, ὄντων ἐπιμορίων· τὸ μὲν ἄρα διὰ πέντε, ἐπειδὴ μεῖζόν ἐστιν, ἡμιόλιον ἂν εἴη, τὸ δὲ διὰ τεσσάρων ἐπίτριτον.

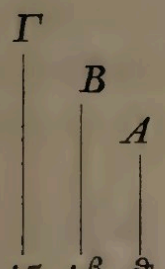
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1. ια΄] V, ιβ΄ M², nec iam sequuntur protaseon numeri in codd. 5. δίς] supra scr. M². 6. ὅν] M² ex ὧν. 13. αὐτό] αὐτῶ M. 25. δέ] Porph., γάρ codd. Post ἐκ insert. τε M⁴.

Menge 1916, printed p. 172; Greek text. Public-domain edition, Internet Archive California scan, PDF p. 230.

XI.

Diatessaron intervallum et diapente utrumque superparticulare est.



nam sit nete synemmenon *A*, mese autem *B*, hypate vero meson *Γ*. *AΓ* igitur intervallum, quod est bis diatessaron, dissonans est; quare non est multiplex. iam quoniam duo intervalla aequalia *AB*, *BΓ* composita totum non efficiunt multiplex [prop. V], ne *AB* quidem multiplex est. et est consonans; ergo superparticulare. eadem demonstratio est etiam in diapente.

XII.

Diapason intervallum duplum est.

nam demonstravimus, multiplex id esse [prop. X]. Itaque aut duplum est aut maius quam duplum. verum quoniam demonstravimus, intervallum duplum ex duobus maximis superparticularibus constare [prop. VI], ita ut, si diapason maius erit quam duplum, non constet ex duobus solis superparticularibus, sed ex pluribus, — constat vero ex duobus intervallis consonantibus, ex diapente et diatessaron, diapason non erit maius quam duplum, ergo duplum.

sed quoniam diapason duplum est, intervallum autem duplum ex duobus maximis superparticularibus constat [prop. VI], etiam diapason ex intervallis sesquealtero et sesquetertio constat, nam haec maxima sunt, constat autem ex diapente et diatessaron, quae sunt superparticularia [prop. XI]. ergo diapente, quia est maius, sesquealterum est, diatessaron vero sesquetertium.

Menge 1916, printed p. 173; Latin translation. Public-domain edition, Internet Archive California scan, PDF p. 231.

φανερὸν δή, ὅτι καὶ τὸ διὰ πέντε καὶ διὰ πασῶν τριπλάσιόν ἐστιν. ἐδείξαμεν γάρ, ὅτι ἐκ διπλασίου διαστήματος καὶ ἡμιολίου τριπλάσιον διάστημα γίνεται, ὥστε καὶ τὸ διὰ πασῶν καὶ τὸ διὰ πέντε τριπλάσιον.

τὸ δὲ δὶς διὰ πασῶν τετραπλάσιόν ἐστιν.

ἀποδέδεικται ἄρα τῶν συμφώνων ἕκαστον, ἐν τίσι λόγοις ἔχει τοὺς περιέχοντας φθόγγους πρὸς ἀλλήλους.

ιγ'.

Λοιπὸν δὴ περὶ τοῦ τονιαίου διαστήματος διελθεῖν, ὅτι ἐστὶν ἐπόγδοον.

ἐμάθομεν γάρ, ὅτι, ἐὰν ἀπὸ ἡμιολίου διαστήματος ἐπίτριτον διάστημα ἀφαιρεθῇ, τὸ λοιπὸν καταλείπεται ἐπόγδοον. ἐὰν δὲ ἀπὸ τοῦ διὰ πέντε τὸ διὰ τεσσάρων ἀφαιρεθῇ, τὸ λοιπὸν τονιαῖόν ἐστι διάστημα· τὸ ἄρα τονιαῖον διάστημά ἐστιν ἐπόγδοον.

ιδ'.

Τὸ διὰ πασῶν ἔλαττον ἢ ἓξ τόνων.

δέδεικται γὰρ τὸ μὲν διὰ πασῶν διπλάσιον, ὁ δὲ τόνος ἐπόγδοος· τὰ δὲ ἓξ ἐπόγδοα διαστήματα μείζονα διαστήματός [ἐστι] διπλασίου. τὸ ἄρα διὰ πασῶν ἔλαττόν ἐστιν ἓξ τόνων.

ιε'.

Τὸ διὰ τεσσάρων ἔλαττον δύο τόνων καὶ ἡμιτονίου, καὶ τὸ διὰ πέντε ἔλαττον τριῶν τόνων καὶ ἡμιτονίου.

ἔστω γὰρ νήτη μὲν διεζευγμένων ὁ Β, παραμέση δὲ ὁ Γ, μέση δὲ ὁ Δ, ὑπάτη δὲ μέσων ὁ Ζ. οὐκοῦν τὸ μὲν

---

6. Post ἄρα add. ὅτι M². ἕκαστον] ex ἕκτον corr. M². 7. λόγοις] -οις om. in fine lin. M. 9. δή] B, δεῖ M. 10. ἐπόγδοον] δ in ras. M. 11. ἐάν] ἐὰν μέν in ras. M³. 17. Post ἔλαττον add. ἐστιν M³.

Menge 1916, printed p. 174; Greek text. Public-domain edition, Internet Archive California scan, PDF p. 232.

iam manifestum est, et etiam diapente et diapason triplum esse. demonstravimus enim, ex intervallo duplo et sesquealtero intervallum fieri triplum [prop. VII], ita ut etiam diapason et diapente sit triplum.

disdiapason quadruplum.

ergo demonstratum est, singula consonantia intervalla in quibus rationibus sonos, quibus continentur, inter se habeant.

XIII.

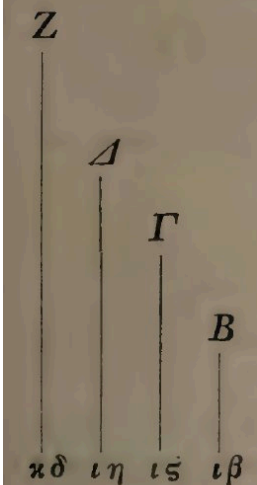
Iam restat, ut de toniaeo intervallo disseramus, esse sesqueoctavum.

nam didicimus, si ab intervallo sesquealtero intervallum sesquetertium auferatur, quod relinquitur, sesqueoctavum esse [prop. VIII]. verum si a diapente diatessaron auferatur, quod relinquitur, toniaeum est intervallum. ergo toniaeum intervallum sesqueoctavum est.

XIV.

diapason minus est quam sex tonorum.

nam demonstratum est, diapason duplum esse [prop. XII], tonum vero sesqueoctavum [prop. XIII]. verum sex intervalla sesqueoctava intervallo duplo sunt maiora [prop. IX]. ergo diapason minus est sex tonis.



XV.

Diatessaron minus est duobus tonis semitonioque et diapente minus est tribus tonis semitonioque.

nam sit nete diezeugmenon B, paramese autem Γ, mese autem Δ, hypate

Lineae fig. in M² aequales sunt; numeri desunt.

Menge 1916, printed p. 175; Latin translation. Public-domain edition, Internet Archive California scan, PDF p. 233.